PrescriptoBot

From Scribbles to Scripts — Instantly

Bishoy Kamel- Habiba Walid - Abdelrahman Walid

CURABOTICS

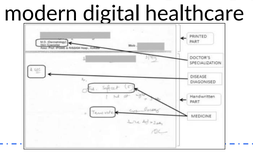
Abstract

In a time where misreading handwritten prescriptions risks patient safety, our AI-powered system extracts, interprets, and schedules medication instructions using **computer vision** and **NLP**. It accurately handles **Arabic-English text**, detects drug names, and automates dosing—boosting healthcare efficiency and reducing human error.

Introduction

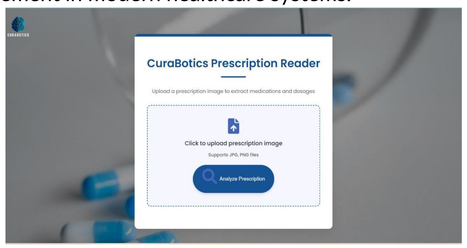
Handwritten prescriptions remain a common yet problematic element in modern healthcare, often leading to confusion, dosage errors, and administrative delays.
Machathon 6.0 challenges us to harness the power of AI to transform this outdated practice.
Our solution aims to digitize handwritten prescriptions using OCR and NLP techniques, handling multilingual content, spelling variations, and ambiguous time instructions. Through intelligent text extraction and scheduling automation, we bridge the gap between traditional practices and modern digital healthcare systems.

Text written by hand



Aim & Objectives

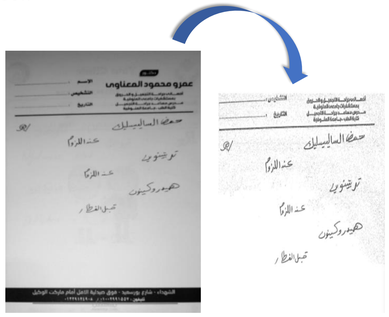
This project aims to develop an AI-based system that digitizes handwritten medical prescriptions to enhance patient safety and healthcare efficiency. The objective is to build a lightweight model capable of accurately extracting and interpreting mixed-language (Arabic-English) handwritten text, identifying medicine names and dosage instructions despite spelling inconsistencies or handwriting noise.
Additionally, the system will automatically convert appointment instructions into structured schedules, providing a clear, user-friendly output.
By combining OCR, NLP, and smart preprocessing techniques, the solution seeks to reduce human error and streamline prescription management in modern healthcare systems.

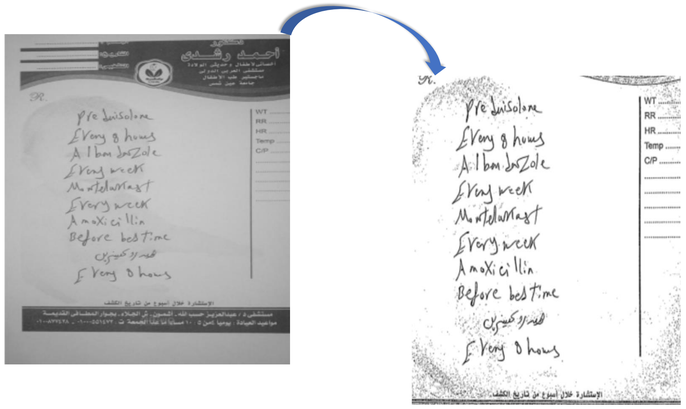


Methodology

To address the challenge of handwritten prescription recognition, our approach was divided into several key stages:.

1.Image Preprocessing: We enhanced the dataset by applying various augmentation techniques such as brightness adjustment, noise reduction, and contrast enhancement to improve the model's robustness. Additionally, we synthetically expanded the dataset to capture more handwriting variations and conditions, which helped reduce overfitting and improve generalization during training





2. Model Exploration

We evaluated several model architectures to find the best fit for handwritten, multilingual prescriptions:

•Custom R-CNN + CTC Loss:
Built from scratch, this model struggled with variable-length lines and inconsistent image sizes, leading to truncated/misaligned outputs.

•ViT + Transformer (Seq2Seq):
The sequence-to-sequence approach had strong theoretical potential but performed poorly on noisy, handwritten data and required long training time.

•TrOCR Variants:

- **Base (pretrained on IAM):** Handled English well but failed with Arabic and mixed-language inputs.
- **Base (trained on prescription data):** Overfitting persisted; still weak on Arabic due to limited capacity and prior exposure.
- **Large TrOCR:** Showed better generalization but demanded heavy resources and still needed diverse Arabic data for optimal performance.

Based on these findings, we refined our approach to strike a balance between **accuracy, efficiency, and real-world usability**.

Results & Analysis

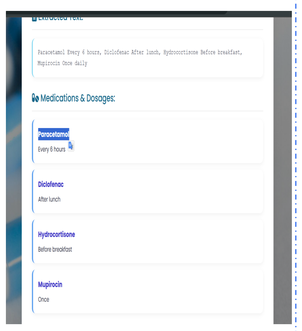
Challenges

•**Handwritten Data:** Variations in handwriting styles made recognition inconsistent and error-prone.
•**Mixed Arabic-English Text:** Multilingual content complicated preprocessing and model predictions.
•**Insufficient Arabic Data:** We created a custom questionnaire to collect real-world Arabic prescriptions.
•**Low Image Quality:** Some data suffered from poor lighting, resolution, and noise.
•**Resource Constraints:** Limited computational power prevented training large models like **Donut Base (fine-tuned)**.

Results

To assess the effectiveness of our handwritten prescription recognition approach, we tested multiple model configurations and preprocessing methods:

1.Image Preprocessing Impact
Augmenting the dataset (e.g., brightness adjustment, noise reduction, contrast enhancement) improved robustness, reduced overfitting, and enhanced generalization. Synthetic variations helped manage handwriting inconsistencies, especially in limited Arabic-script data.



2. Model Performance Comparison

Model Variant	Language Support	Overfitting	Accuracy (Mixed Scripts)	Training Time	Deployment Feasibility
Custom R-CNN + CTC Loss	Weak Arabic support	High	Low	Moderate	Moderate
ViT + Transformer (Seq2Seq)	Moderate	Moderate	Moderate	High	Low
TrOCR Base (IAM Pretrained)	English only	Very High	Very Low	Low	High
TrOCR Base (Prescription -trained)	Weak Arabic generalization	High	Low	Moderate	Moderate
TrOCR Large (Fine-tuned)	Strong multilingual	Reduced	Best	High	Low

•TrOCR Large outperformed other models on mixed Arabic-English prescriptions, requiring careful fine-tuning to prevent overfitting. Despite its higher computational cost, it achieved the best WER and CER balance.